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Design Modifications for Energy Conservation of Sponge Iron Plants

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During the operation in the coal based sponge iron plant, a tremendous amount of heat is generated and significant part of this heat, associated with the waste gas, remains unutilized. It appears worth interesting to modify the process that facilitates the integration of heat available with the waste gas. In the present paper, for the utilization of heat of waste gas, two modifications, namely, case-1 and case-2, are proposed based on preheating of inlet streams. In case-1, preheating of feed material is considered, whereas in case-2, preheating of feed material as well as air is accounted. These cases are then compared with the existing waste heat recovery system of the plant based on coal consumption, operating and capital costs, profit, and payback period. It is found that both cases are better than the existing heat recovery system of the plant. However, case-2 is selected as the best heat recovery option. In comparison to the existing system, case-2 reduces coal and water consumption by 30.5% and 72.6%, respectively. Further, case-2 releases minimum waste gas to the atmosphere that makes the process environment friendly.
[DOI: 10.1115/1.4003506]

Keywords: sponge iron plant, heat integration, coal consumption, economic analysis

1 Introduction

Sponge iron is not soft and spongy, as the name might appear, but a metallic mass with honeycomb structure all over the surface and bulk. Since past decade, sponge iron has emerged as a raw material for manufacturing steel. Seeing its tremendous potential, the government has recognized sponge iron as a vital sector for the growth of Indian steel industry. Again, India is fortunate to have high reserves of good quality iron ore and large resources of noncoking coal used in making sponge iron. The average moisture contents in coal and iron ore are about 13% and 2%, respectively [1].

Sponge iron manufacturing units looked profitable since the beginning of the 1990s. However, with the increase in the input cost of raw material and decrease in selling price, it finds difficulty to survive. The lack of infusion of modern technology has also added further dimensions to the impasse. In the present situation, even though the condition has improved considerably, sponge iron industry passes through several problems, such as lack of proper integration of heat energy, obsolete technology, and nonoptimal operation of equipment, which are making it a less profitable venture. Thus, there exists a large scope to apply modern technology to this industry and to make it more competitive by cutting down its internal losses. The present study utilizes the above opportunity effectively and puts forward suggestions to improve the energy effectiveness of the industry through design modifications.

Many investigators, such as Agrawal et al. [1,2], Rani Devi and Mazumder [3], and Misra and Ipicol [4], considered the process of sponge iron manufacturing and suggested improvement in that. However, as the present paper is based on energy savings of sponge iron plant, literature, focused on this aspect, is discussed here.

Jena et al. [5] focused on different aspects such as waste gas composition, dust loss, air requirement, and efficiency of the process. Due to chemical reactions and combustion, the heat generated inside the kiln is 174.28 GJ/h and heat value of coal input is 323.2 GJ/h, resulting in a thermal efficiency of the process to be 46%. They also showed that 33% of heat generated in the kiln is lost with waste gas. The authors suggested that heat in waste gas may be recovered by putting a waste heat recovery system for generation of power.

Bandyopadhyay et al. [6] discussed that coal is the important raw material for sponge iron making and revealed that if very high volatile matter (VM) coals are used, most of the VM is lost at the kiln mouth when coal encounters counterflowing hot waste gases. Agarwal and Sood [7] found that out of the total heat generated from coal, only 35% is used in reduction and balance goes as waste. Elsenheimer and Serbent [8] also mentioned that the energy contained in waste gas in form of sensible or chemical energy amounts up to 40%. They proposed a number of options to recover the energy but did not show how much energy can be saved by installing these options and what their practical implications are.

Hajidavalloo and Alagheband [9] studied the thermal analysis of sponge iron preheating using waste energy of electric arc furnace (EAF). To improve performance of EAF, a new initiative technique has been introduced, in which sponge iron particles are preheated before entering the furnace. Based on the simulation results, it is found that the energy consumption in the EAF and productivity are reduced up to 14% and 13%, respectively.

Biswas et al. [10] presented the potential for energy savings in various capacity sponge iron plants by estimating heat transfer rates using temperature data within the kiln. Air jet seal instead of mechanical seal arrangements at specific locations in the rotary kiln and cooler were proposed and thus power consumption was reduced.

An energy survey was made by Eriksson and Larsson [11] on a sponge iron plant in order to identify the largest energy losses and find ways to increase efficiency. They suggested that largest improvements in energy efficiency would be obtained by building a new kiln. Recently, Mignard and Pritchard [12] reviewed the sponge iron process for the storage and transmission of remotely generated marine energy.

Literature indicates that power consumption for Indian sponge iron industries ranges from 45 kW h/t to 130 kW h/t [13]. The energy requirement for rotary kiln ranges between 14.63 GJ and 20.9 GJ per t sponge iron. However, during the operation in coal based sponge iron plant, tremendous amount of heat is generated that remains unutilized. It is therefore understandable that energy questions will always be important in these industries and, hence, one should integrate heat in these industries effectively. Thus, in the present paper, an approach is discussed to utilize heat of waste gas for preheating the feed materials as well as air. As a result,

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Contributed by the Heat Transfer Division of ASME for publication in the JOURNAL OF THERMAL SCIENCE AND ENGINEERING APPLICATIONS. Manuscript received March 16, 2010; final manuscript received January 19, 2011; published online March 10, 2011. Assoc. Editor: Chenn Zhou.

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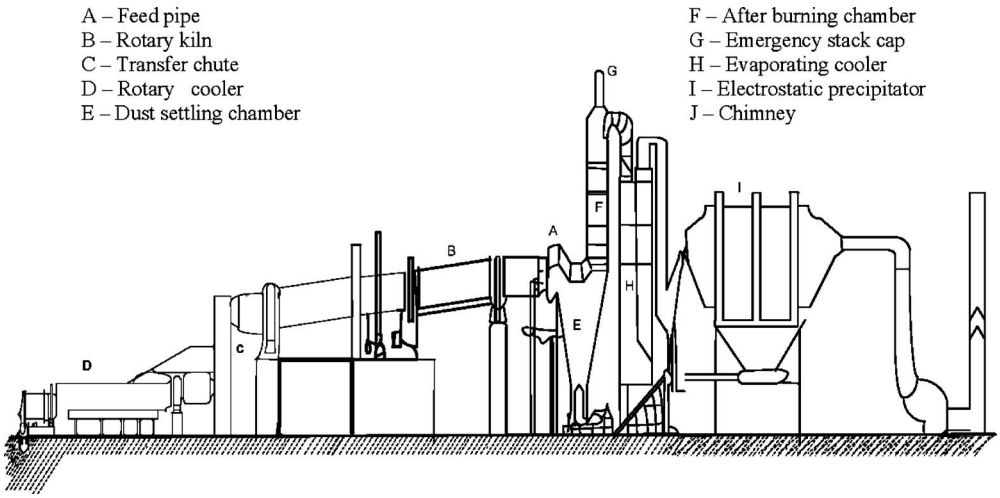


Fig. 1 Longitudinal section of sponge iron plant

reduced amount of coal is predicted. The equipments required for modification are designed and their economic analyses are also presented.

2 Conventional Coal Based Sponge Iron Plant: A Brief Description

The longitudinal section of typical coal based sponge iron plant is shown in Fig. 1. The sized iron ore, feed coal, and dolomite are fed to the rotary kiln through kiln feed pipe. A separate conveyor collects different size fractions of coal for injection into the kiln from the discharge end side. All along the kiln length, air is injected through eight air fans in the conventional 500 t/day sponge iron plant, each of them are adjusted separately. Further, air is injected at the kiln outlet by a central burner pipe, which, during normal operation, serves as process air inlet, as shown in Fig. 2.

Inside kiln is lined with refractory and supported on three piers called support rollers with an inclination of 2.5 deg. Rotation is given by the girth gear. Due to its inclination and rotation, the material in the kiln moves along the axis. Both ends of the rotary kiln is provided by the mechanical air sealing so that no ingress of air takes place. As charge moves through the kiln, it is heated by gases that flow in opposite direction. At the inlet side of the kiln, blowing of controlled amount of air into the material charge increases material temperature by direct combustion of volatiles within the charge.

The first section, approximately half of the kiln, is called the preheating zone, where iron ore, coal, and dolomite are heated to reaction temperatures and the second half is the reduction zone, as shown in Fig. 3. In the preheating zone, moisture of the materials is driven off along waste gas. The volatiles of coal are, to one part, escaping into the gas space above the material bed and, to other part, directly burned within material charge, where they are used

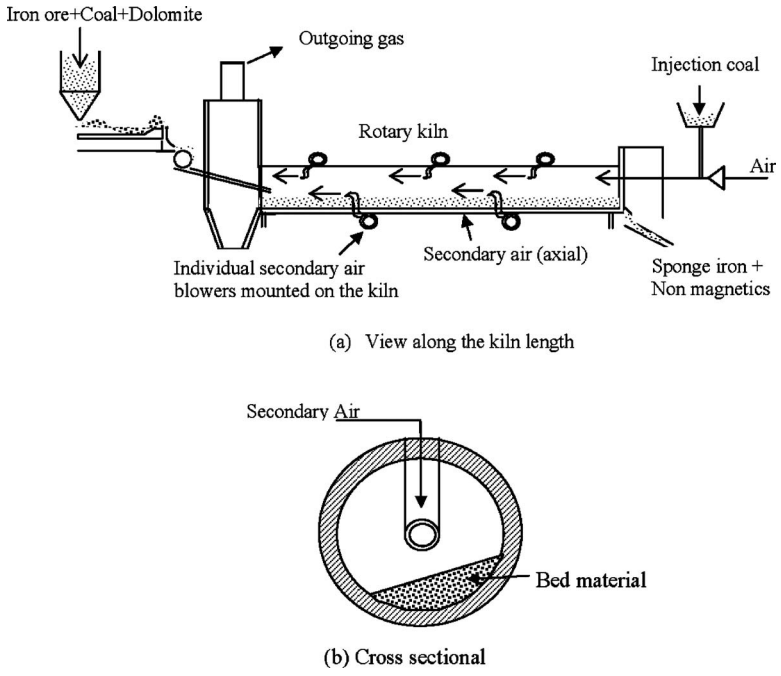


Fig. 2 Normal injection of air into the kiln

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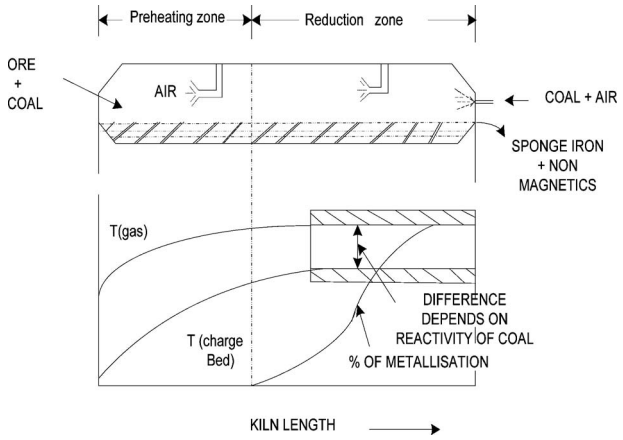


Fig. 3 Principle of counterkiln operation

Table 1 Mass balance around main components of the plant on per hour basis

Rotary kiln			
Input		Output	
Component	Mass *t*	Component	Mass *t*
Iron ore	30	Sponge iron	18.6
Feed coal	15.875	Waste gas	103.426
Dolomite	0.5	Total	122.026
Slinger coal	5.562		
Air	70.089		
Total	122.026		
ABC			
Waste gas	103.426	Dust	5.4
Air	13.35	Waste gas	128.376
Water	17	Total	133.776
Total	133.776		

for direct heat up. The volatiles in the gas space are burned by air and thus supply main energy for heating up the kiln charge. In the reduction zone, the main portion of oxygen contained in the iron ore is removed, leaving metallic iron and very few iron oxides behind. The difference of the temperature of free gas T(gas) and bed material T(charge bed) is shown in Fig. 3 along the length of kiln. It is responsible for heating bed charge and metallization of product. Plot of the percent of metallization is shown in Fig. 3. It indicates that reduction of iron ore starts in the reduction zone and metallization of product is achieved at the discharge end of the kiln. The reduced product from the kiln coming through the transfer chute is indirectly cooled in a rotary cooler.

Waste gas in the kiln flows in opposite direction to the feed materials. The flow is maintained by an induction fan mounted before the chimney. The waste gas coming out from the rotary kin is processed by different equipments before it leaves to the open atmosphere. The direction of flow of waste gas after rotary kiln is shown in Fig. 4.

Generally, waste gas consists of N_2, CO_2, CO, H_2, H_2O, O_2, and CH_4 with compositions of 59.69%, 35.91%, 0.32%, 0.17%, 3.7%, 0.08%, and 0.13%, respectively. It comes out of the kiln at a temperature about 900°C and is then taken to after burner chamber (ABC) and dust settling chamber (DSC) that is located beneath the ABC. DSC reduces waste gas velocity, removes large dust particles by gravity, retards pressure fluctuation, and achieves uniformity of waste gases with regard to temperature and concentration of combustible. At the end of DSC, the waste gas changes its direction and move upward into combustion area of ABC. Here, combustibles are mixed with fresh excess air and burnt completely to acquire temperature in a range between 950°C and 1050°C approximately.

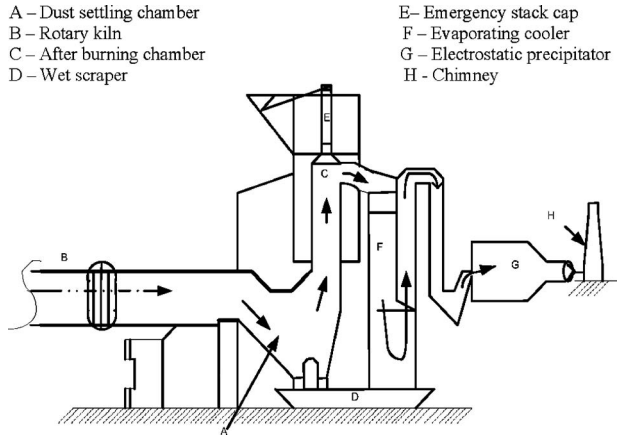


Fig. 4 Kiln with waste gas processing equipments and its flow direction

The temperature of waste gas needs to be brought down to a workable limit for downstream equipments to remove dust particles and toxic components. Thus, water is sprayed to reduce the temperature of waste gas in the ABC. For this purpose, ten numbers of water gun is fitted along different heights. The water guns hold a nozzle at the discharge end for atomization. The pressurized water droplets coming out from the guns fall on waste gas, resulting in an increase in the weight of dust particles that helps it to settle down.

During the plant operation, there are some unstable conditions and feeding in the kiln is totally stopped. For such conditions, the emergency cap, located at the top of ABC, is operated and waste gases are allowed to escape to open atmosphere.

The evaporating cooler (EC) is connected with the ABC, as shown in Fig. 4. Here, waste gas is quenched with water. Eight to ten numbers of water guns are provided at top position around the EC at same height. Water coming out from these is sprayed to reduce the temperature of waste gas at desired level. The bottom part of the DSC, EC, and ABC is attached with the wet scraper.

Waste gas exiting the EC enters the electrostatic precipitator (ESP) for final purification. The desired temperature of waste gas is to be maintained below 250°C. ESP is considered to be the most effective dust collector in industries due to low pressure drop, low sensitivity at high temperature and aggressive gases, high collection efficiency, and low maintenance. ESP exit is connected to the chimney. Induced direct fan is placed before the chimney. After ESP, filtered waste gas goes to the atmosphere.

Mass balance around the main equipment, such as rotary kiln and ABC of a typical sponge iron plant of 500 t/day capacity, carried out on per hour basis, is shown in Table 1. The energy balance is done in the similar lines, as shown by Jena et al. [5]. The schematic diagram for conventional plant is shown in Fig. 5. The actual coal consumption for the conventional plant is 21.437 t/h, in which 15.875 t/h is used as feed coal and the rest are used as slinger coal.

2.1 Existing Heat Recovery System in the Conventional Plant. For utilization of heat of waste gas exiting from ABC, a heat recovery boiler system is installed in the conventional plant for generating power. The plant data show that this option generates 2.6 MW of power. The power may be exported with *Rs* 3.5 kW h. It gives *Rs* 728 lakh/yr profit for exporting power.

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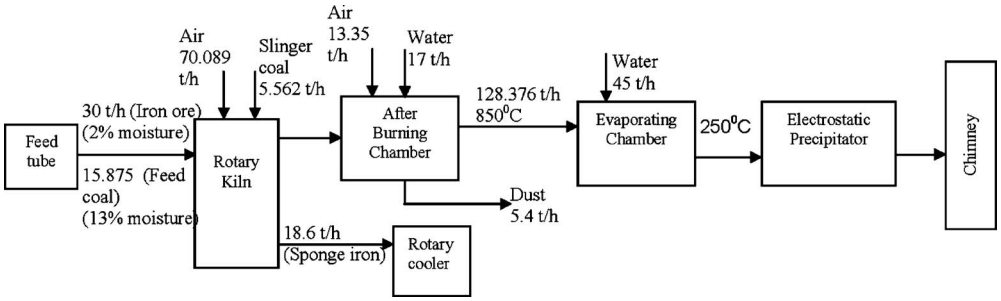


Fig. 5 Schematic diagram of the conventional process

However, the capital cost of this system is *Rs* 765 lakh. These costs are directly taken from the typical sponge iron plant of 500 *t*/day capacity.

3 Modification in the Conventional Plant for Utilization of Waste Heat

The process, used in sponge iron industry, produces a significant amount of heat that remains unutilized. One of such example is the heat associated with waste gas. In the present work, heat of waste gas is utilized for preheating of inlet streams such as feed material and air. For this purpose, two modifications, namely, case-1 and case-2, are proposed. In case-1, preheating of feed material is considered, whereas in case-2, preheating of feed material as well as air is accounted. In other words, cases-1 and -2 are selected based on the increase in utilization of heat of waste gas. For these modifications, variation of physical properties of waste gas, feed material, and air is assumed to be negligible.

3.1 Preheating of Feed Material. The possible approach to recover heat of waste gas is to preheat the feed to rotary kiln. The feed materials to kiln are iron ore and coal. For preheating, a rotary drier is used as a heat exchanger. Some part of waste gas, which exits from ABC, is fed to the rotary drier at 750°C and taken out at 250°C.

Preheating of feed is used to remove the moisture of coal as well as iron ore. For calculation purpose, it is considered that 30 *t*/h iron ore requires 15.875 *t*/h feed coal. The computation of actual coal consumption after the preheating of feed material is shown in Appendix A. This modification is referred as case-1.

3.1.1 Design of Rotary Drier. To design rotary drier, heat loss surface area is to be calculated. For this purpose, drop in temperature of waste gas in the drier due to loss of heat through shell is required, which is computed using energy balance, as shown below.

Enthalpy change of waste gas in the rotary drier=required energy to heat the moisture from ambient temperature to 100°C +required energy to vaporize the moisture at 100°C+required heat to heat the water vapor to outgoing waste gas temperature, i.e., T_{fo}+required energy to heat the feed coal (moisture free) from ambient temperature to outlet temperature, i.e., T_{co} +required energy to heat the iron ore (moisture free) from ambient temperature to outlet temperature, i.e., T_{co}+heat lost to the surroundings through shell of the drier.

Detailed computation of different terms of energy balance is shown in Appendix B. It gives total heat loss to surroundings through shell of drier and is 5.02×10^5 W. Drop in temperature of waste gas in the drier due to loss of heat is found to be 25°C, which took place at different sections of the drier. For accuracy, heat transfer surface area of the drier for the temperature drop of 25°C of waste gas is calculated using Eq. (1) [14].

$$T_{fo} = T_{fi}\exp\left(-\frac{UA}{m_f c_f}\right) + T_a\left[1-\exp\left(-\frac{UA}{m_f c_f}\right)\right] \tag{1}$$

where U is the overall heat loss coefficient and is given by

$$U = \left[\frac{1}{h_{gl}} + \frac{t_{dp}}{k_{dp}} + \frac{1}{h_{\text{roa}} + \frac{1}{h_{\text{coa}}}}\right]^{-1} \tag{2}$$

Considering m_f, c_f, and U as 18.15 kg/s, 1.14 kJ/kg K, and 7.037 W/m^2 K, respectively, heat loss surface area of the drier is calculated and presented in Table 2. Average drier heat loss surface area is calculated to be 174.3 m^2. Considering the inside diameter of drier as 3 m [15], the length of the drier is found as 18.5 m.

The most important design parameter that affects the performance of the rotary drier is the residence time τ of the feed materials. For its computation, an empirical correlation, proposed by Friedman and Marshall, Jr. [16] and shown through Eq. (3), is used

$$\tau = \left(\frac{0.3L}{\tan\alpha N^{0.9} D} - \frac{16.9LG}{d_p^{0.5} F}\right) \tag{3}$$

The rotational speed and slope of the drier are taken as 3.95 rpm and 1.03 deg, respectively. It gives residence time as 27.5 min. The amount of moisture removed increased with the increase

Table 2 Heat loss surface area of the drier for the temperature drop of 25°C

T_{fi} (°C)	T_{fo} (°C)	A (m^2)
750	725	103.9
725	700	107.7
700	675	111.8
675	650	116.2
650	625	121
625	600	126.2
600	575	131.9
575	550	138
550	525	144.9
525	500	152.4
500	475	160.7
475	450	170
450	425	180.4
425	400	192.2
400	375	205.7
375	350	221.2
350	325	239.2
325	300	260.4
300	275	285.7
275	250	316.5

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in residence time. However, if residence time is more than the required, then the rate of dusting is increased [17]. Thus, residence time of 27.5 min should be maintained in the drier to remove moisture.

3.1.2 Duct Carrying Waste Gas. The role of duct is to convey the waste gas from the top of the ABC to the rotary drier at desired temperature. The shape of the duct is cylindrical to reduce the surface area results in minimum heat loss during conveying the waste gas. The length of the duct is 43 m. The outside and inside diameters are 1.6 m and 1.2 m, respectively. The detailed design of the duct is shown in the work of Prasad et al. [14]. This design gives the temperature at the outlet of the duct as 750°C. The casing of the duct is of mild steel and inside it, 200 mm thickness of the insulating refractory brick is placed. The selection of the brick is carried out based on thermal conductivity, which is computed using the concept of heat loss through composite pipe [14]. In the present work, refractory brick of thermal conductivity 114.36 kJ/mh °C is used. Magnesia fused or magnesia brick as refractory for the waste gas carrying duct is appropriate in this case [18].

3.2 Preheating of Air. In the conventional process, waste gas of significant amount exits from the ABC at considerably higher temperature. A part of it is utilized in the preheating of feed, as illustrated in Sec. 3.1. Remaining part of waste gas can be used to preheat air. In the conventional process, air is preheated inside the kiln up to the reaction temperature before reduction takes place in the kiln. This heat is supplied by combustion of coal. Due to preheating of air outside the kiln using heat of waste gas, the heat supplied by combustion of coal for this purpose is reduced, which, consequently, reduces the amount of coal required. Another benefit of this modification is that waste gas achieves T_{fo} through preheating the air and thus water requirement in EC will be eliminated. This modification is named as case-2. It is over and above case-1 as in this case, along with the preheating of feed materials, air preheating is also considered.

3.2.1 Gas-Gas Heat Exchanger. For preheating of air using waste gas, a shell and tube heat exchanger (STHE) with gas-gas combination is used. The detailed design of shell and tube heat exchanger is taken from Ref. [19]. The computation of heat transfer area is found using heat load and heat transfer coefficient. The value of heat transfer coefficient is taken as 50 W/m^2 °C [19]. For carrying waste gas ducts to STHE are used in a similar manner as discussed in Sec. 3.1.2.

4 Economic Analysis

For the proposed modifications, case-1 and case-2, rotary drier, STHE, and ducts are installed to preheat the feed material and air to kiln. The method for computing capital and operating costs associated with these modifications are discussed here. However, detailed computation of capital and operating costs for case-1 is shown in Appendix C as sample calculation.

4.1 Capital Cost

4.1.1 Cost of Duct From ABC Outlet to Rotary Drier. The duct is made of mild steel and inside it, insulating bricks are used. The cost of duct depends on the weight that is computed based on the dimensions of the duct. Using the weight of duct and cost of unit weight of mild steel, the total cost of duct is computed. Further, it includes the cost of insulating bricks. In a similar manner, the costs of other ducts are computed.

4.1.2 Cost of Rotary Drier With All Accessories and Shell and Tube Heat Exchanger. The cost of rotary drier includes the costs of three phase induction motor used to drive the drier, gear box, installation, girth gears, support rollers, drier mounted instruments, weigh feeders, conveyors, and vibrofeeders.

4.1.3 Cost of Induced Direct Fan, Motor, Bearing, Coupling, and Electrical Works. Based on the market values cost of induced direct fan, three phase motor, bearing, coupling, electrical works, and installation are computed.

4.1.4 Cost of STHE. Cost of STHE is computed based on the heat transfer area using the following expression [20]:

$$\text{capital cost} = 1{,}368{,}000 + 34{,}200A^{0.81} \tag{4}$$

4.2 Operating Cost. The operating cost of the modification depends on the amount of coal utilized in the process. It does not include the cost of iron ore as it is fixed for two modifications, case-1 and case-2.

4.3 Profit and Payback Period. Due to the preheating of feed material and air coal, requirement of the process is reduced. Thus, savings due to reduction in coal consumption gives the total profit of the process. Based on capital cost, obtained in Sec. 4.1, and profit, payback period of the modification is computed.

5 Coal Consumption for Overall Process

The coal consumption of sponge iron production process depends on the total energy required in the process, which is supplied by the combustion of coal. Thus, the present work requires the computation of energy and coal consumption.

Total energy requirement of the conventional process depends on (i) heat gained by incoming air and feed materials, (ii) heat required for the endothermic reaction process, (iii) heat lost through the rotary kiln wall, (iv) heat gained by coal, and (v) heat required by moisture of the feed material to evaporate. The estimation of total energy required for the conventional process consists of the following steps.

- Heat gained by the inlet air and bed of material: In the preheating zone, inlet air and iron ore is heated up to reaction temperature and then reduction takes place in the respective zone. The sensible heat gained by air and ore is supplied by combustion of coal and computed using the following equations:

$$Q_a = M_a C_a (t_p - t_a)$$

$$Q_s = M_s C_s (t_p - t_a)$$

- Heat required for the reduction process to continue [10]: The stoichiometric equations responsible for the reduction of iron ore are

$$1/2\ \ Fe_2O_3 + 3/2\ \ CO = Fe + 3/2\ \ CO_2 - 12{,}866\ \ \text{kJ/kmol} \tag{5}$$

$$1/3\ \ Fe_3O_4 + 4/3\ \ CO = Fe + 4/3\ \ CO_2 - 3904\ \ \text{kJ/kmol} \tag{6}$$

$$FeO + CO = Fe + CO_2 - 16{,}108\ \ \text{kJ/kmol} \tag{7}$$

$$FeO + C = Fe + CO + 156{,}482\ \ \text{kJ/kmol} \tag{8}$$

Equation (8) indicates the heat required for the desired rate of reduction of iron oxide to iron. CO generated by Eq. (8) is utilized either for further reduction of iron ore through Eqs. (5)–(7) or for combustion ($2\ CO + O_2 \rightarrow 2\ CO_2$). It is revealed from Eq. (8) that 56 kg of sponge iron requires approximately 12 kg of carbon and 156,482 kJ/kmol of endothermic heat. Therefore, for the endothermic process, heat (kJ/h) requirement is computed as

$$Q_p = (156{,}482/56)M_s$$

- Heat lost through the kiln wall and refractory [5]: The actual heat loss through the wall includes lost through the kiln

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Table 3 Economic analyses of cases-1 and -2

Cases	Operating cost (lakh/yr)		Fixed cost (lakh)		Total capital cost (lakh)	Total profit (lakh/yr)	Payback period (month)
	Commodity	Cost	Item	Cost			
1	Coal	3243.3	Drier	290.5	429.2	623.6	8.37
			Ducts	33.4			
			ID fan	84.3			
			Others	21			
2	Coal	2684.9	STHE	44.6	674.1	1180.6	6.95
			Drier	290.5			
			Ducts	128.3			
			ID fan	168.7			
			Others	42			

shell, inlet and outlet hoods, and post combustion chamber and inlet area of the cooler. Therefore, total heat loss is assumed as twice as that of the kiln.

$$Q_{\text{loss}} = 2 \times (\pi DL \times h_r)$$

where h_r is the heat transfer coefficient.

- Heat gained by coal: The coal is also required to be preheated up to the reaction temperature and sensible heat involved for this is shown as

$$Q_c = m_c C_c (t_p - t_a)$$

- Heat required for removing the moisture of feed material: The average moisture contents in coal and iron ore are 13% and 2% [1], respectively, by weight in Indian conditions and the heat required to remove it is computed as

$$Q_m = m_m \times \lambda$$

$$Q_m = (0.02 m_s + 0.13 m_c) \times \lambda$$

It is considered as 70% of the total fixed carbon available in noncoking coal burns completely to give out the heat and the remaining 30% do not burn. The major fraction of this unburnt carbon is lost to the atmosphere through waste gas as smoke, and the rest is discharged with sponge iron from the rotary kiln. The final empirical relation for estimating coal requirement is

$$Q_s + Q_a + Q_p + Q_{\text{loss}} + Q_c + Q_m = m_c \text{NHV}(0.7) \tag{9}$$

or

$$Q_s + Q_a + Q_p + Q_{\text{loss}} + 0.02 m_s \times \lambda = m_c [\text{NHV}(0.7) - C_c (T_p - T_i) - 0.13 \times \lambda] \tag{10}$$

The value of m_c can be predicted from Eq. (10), considering the NHV of coal as 22,930.5 kJ/kg. The property data of air, ore, and coal are constant and taken from literature [21,22].

6 Results and Discussion

In the present work, two modifications are proposed for heat recovery in the conventional process of sponge iron production and named as case-1 and case-2. These cases are discussed here in detail.

6.1 Case-1. In this case, waste gas is used to preheat the feed material (iron ore and coal) to kiln up to 110°C to remove the moisture. Consequently, coal consumption is reduced to 17.98 t/h. This value is obtained from Eq. (10). In case-1, heating of feed material from 110°C to 1020°C is done inside the kiln by the heat generated from combustion of coal instead of from 30°C to 1020°C as in the conventional plant. Thus, using waste gas heat, coal consumption can be reduced by 16.1% for the overall process. In this case, waste gas of amount 65.34 t/h is utilized, which is available at 850°C after ABC. First, it moves through the duct of 43 m and acquires a temperature of 750°C. After preheating the feed, waste gas cools down to 250°C, which directly enters the ESP through a duct of 20 m. For this purpose, a rotary drier of length and diameter equal to 18.5 m and 3 m, respectively, is used. The cost of rotary drier is *Rs* 290.5 lakh. The combined cost of two ducts, first from ABC outlet to rotary drier and second from rotary drier to ESP input, is found as *Rs* 33.37 lakh. For the movement of waste gas in these ducts, an induced direct fan is required, which costs around *Rs* 84.3 lakh. To install ducts and rotary drier *Rs*, 21 lakh is required for concreting and erection work. Detailed computation of these costs is shown in Appendix C, which gives the total capital cost as *Rs* 429.2 lakh.

In fact, as coal consumption reduces, air requirement for combustion of coal also decreases. The reduced amount of air is computed based on the ratio of air to coal consumption in the conventional process. For case-1, air requirement is found as 69.98 t/h. Based on mass balance around the rotary kiln and ABC, total waste gas exit from ABC is found as 111.46 t/h. From this value, 65.34 t/h is used to preheat the feed material and the rest, amount of gas, i.e., 46.12 t/h, enters EC. In the present case, load of EC is reduced by 64%, which is the additional advantage of the present modification.

Case-1 reduces the coal consumption by 16.1%, which gives a profit of *Rs* 1.7 lakh/day in terms of saving of coal. Thus, total payback period for case-1 is found as 8.37 months. The summary of economic analysis for case-1 is shown in Table 3. The flow diagram for case-1 is shown in Fig. 6. On the other hand, it can be seen that due to preheating of feed, the intake capacity of the rotary kiln can be increased. If iron ore feed rate to kiln is increased by 10%, feed coal is decreased by 14%. This process increases the production of sponge iron by 10%. Thus, payback period for this case is predicted as 6.82 months as profit is increased by decreasing the cost of feed coal and increasing the production of sponge iron.

6.2 Case-2. In this case, along with the preheating of feed material, air preheating is also considered. Waste gas is used for this purpose. In case-2, waste gas of amount 46.12 t/h, which entered EC in case-1, goes to the air preheater at 850°C. After heating, the air waste gas cools down to 250°C. Due to this exchange, air of amount 69.98 t/h is preheated.

The present problem is somewhat different than case-1 in the sense that coal and air is preheated and at the same time it is considered as the source of heat in the process. Here, coal, along with air, burns and provides the necessary heat. Due to the dual nature of coal and air, evaluation of its amount necessitates trial and error computation technique.

The problem associated with the above duality can be better understood by the following example: Initially, air of amount 69.98 t/h is preheated up to 466.6°C as waste gas of amount 46.12 t/h has sufficient heat to preheat air up to 466.6°C. When these conditions, along with the preheating of feed material, are

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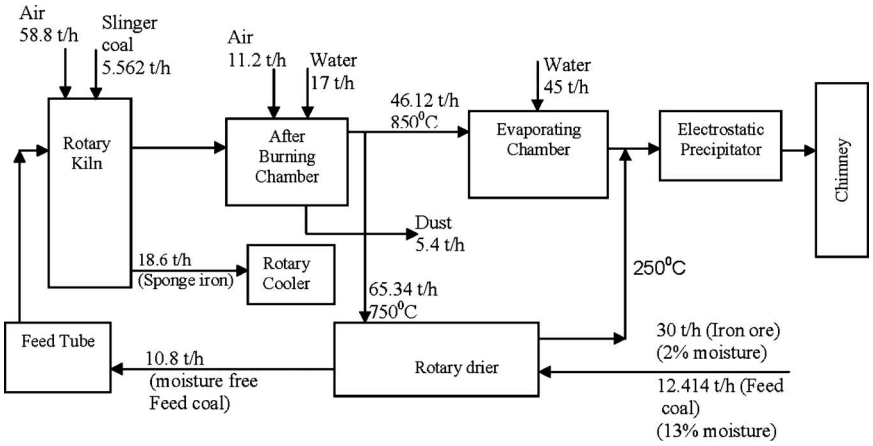


Fig. 6 Schematic diagram of modified process for case-1

used in Eq. (10), coal consumption is found as 14.225 t/h. The amount of air depends on coal consumption, and so, air requirement varies with change in the amount of coal and thus the amount of air corresponds to 14.225 t/h of coal and is found as 49.439 t/h. This new value of coal and air is used for overall mass balance, which gives 90.9 t/h of waste gas. From this amount, 65.34 t/h is used to preheat the feed material and 25.5 t/h is employed to preheat air of amount 49.439 t/h. Now, air can be preheated only up to 358.6°C instead of 466.6°C. Based on this trial and error method, air can be preheated up to 386.7°C. Finally, the computed amount of coal comes out to be 14.884 t/h, which is 30.6% less than the conventional process. For case-2, rotary drier and STHE are employed for the preheating of feed material and air, respectively. The cost of rotary drier is computed in a similar manner that is shown for case-1. However, the cost of STHE is predicted as *Rs* 44.6 lakh using Eq. (4). The heat transfer area of this heat exchanger is found as 260.3 m^2. This modification requires three ducts. The first two are similar to that for case-1 and a third duct of 85 m is required to carry the hot air up to the rotary kiln inlet. In this duct, an insulation of very low thermal conductivity material, such as ceramic fiber, is used, and so, loss of heat through this duct is not significant. While moving through 85 m duct, only 1 deg drop in temperature of air is observed. Now, air enters the kiln at 385.7 deg. The damper inside the blower is made of stainless steel that can withstand with such a high temperature of air. Thus, preheating of air up to 385.7 deg did not cause any difficulty in operation. Considering the inlet temperature of air as 385.7 deg, the coal consumption is found as 14.892 t/h, which is 30.5% less in comparison to the conventional process. To carry the waste gas up to air, preheater duct is not required as the flow of waste gas is in the natural direction, as shown in Fig. 4, and it is maintained by an induction fan mounted at the chimney.

In the present case, water requirement in EC is eliminated, which is the additional advantage of the present modification, as case-2 reduces the coal consumption by 30.5%, which gives a profit of *Rs* 3.23 lakh/day. Thus, total payback period for case-2 is found as 6.95 months. The summary of economic analysis for case-2 is shown in Table 3. The flow diagram for case-2 is shown in Fig. 7.

6.3 Comparison of Case-1, Case-2, and Existing Heat Recovery System of Plant. The comparison of different heat recovery options, such as existing heat recovery system, case-1, and case-2, are shown in Table 4. It indicates that both cases are better than the existing heat recovery system of the plant. Moreover, it is reported that in Indian scenario, generation of power from waste heat is not successful with respect to its operation, stability, and quality of steam [7]. Based on payback period and coal consumption, case-2 is selected as the best heat recovery option.

The reason why case-2 is a better modification in comparison to case-1 is illustrated as in the present sponge iron process, feed material and air first reach from 30°C to reaction temperature, which is 1020°C and then reduction of iron ore takes place to produce sponge iron. To acquire feed material and air, the temperature of 1020°C total heat is supplied by combustion of coal.

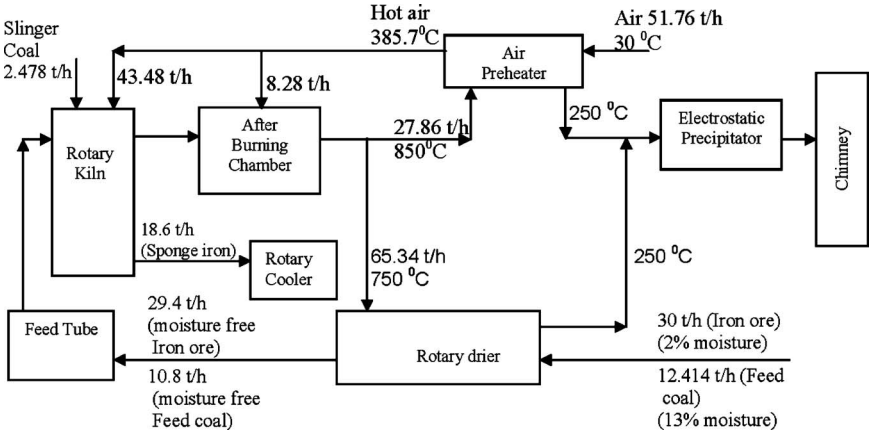


Fig. 7 Schematic diagram of modified process for case-2

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Table 4 Comparison of different heat recovery options

Cases	Operating cost: Coal t/h	Operating cost: Cost (lakh/yr)	Total capital cost (lakh)	Total profit (lakh/yr)	Percent reduction of water consumption	Payback period (month)
Existing	21.437	3866.9	765	728	72.6	12.6
1	17.98	3243.3	429.2	623.6	46.5	8.37
2	14.892	2684.9	588.2	1180.6	72.6	6.95

Cases	Input temperature: Feed material	Input temperature: Air	Input parameter: t of coal/t of sponge iron	Output parameter: Amount of air t/h	Output parameter: Productivity t	Output parameter: Waste gas generated t/h
Existing	30	30	1.15	83.33	18.6	128.376
1	110	30	0.97	70	18.6	111.46
2	110	385.7	0.8	51.76	18.6	93.203

In the present work, certain portion of total heat is provided by the heat available with waste gas through cases-1 and -2 and remaining portion of total heat is supplied by combustion of coal. Case-1 includes the preheating of feed material up to 110°C, which recovers 3297 kW of heat from waste gas. However, for case-2, total heat recovery from waste gas is of 8577.6 kW for the preheating of feed material and air up to 110°C and 385.7°C, respectively. Thus, case-1 and case-2 save 3297 kW and 8577.6 kW, respectively, of total heat of combustion supplied to the conventional process.

The comparison of key input and output parameters for cases-1 and -2 is also shown in Table 4, which indicates the inlet temperatures of feed material and air when these enter the rotary kiln. As inlet temperature of input streams increases, amount of coal requirement per ton production of sponge iron decreases due to the fact that less heat is required for the input stream to reach up to the reaction temperature of 1020°C.

The above analysis clearly indicates that case-2 is the better heat recovery option and consumes less amount of coal in comparison to case-1. One direct benefit of case-2 is that it releases minimum waste gas to the atmosphere, as shown in Table 4, which makes the sponge iron production process environment friendly. In Table 4, percent reduction of water consumption is also presented, which is computed based on the total water consumption in ABC and EC.

Moreover, preheating of feed materials gives dry coal as fuel, which increases the kiln efficiency and fuel thermal value. Much of the combustion heat of coal that could be used in the process is expended in moisture removal. Further, preheating in the drier enhances the intake capacity of rotary kiln, which increases iron ore feed rate at the lower requirement of coal. Consequently, the productivity of the plant is increased. Preheating also allows the effective control on the uniformity of feed materials as moisture contents varies seasonally and depends on the quality.

Preheated zone of the kiln increases if preheating of feed materials and air take place inside the kiln as preheating time increases. That leads to operate the kiln at relatively higher temperature 1170°C in the reduction zone for increasing the heat transfer [10]. Higher temperature for sponge iron making leads to operational problems such as localized heating and liquid phase formation. Preheating of the feed materials and air maintains maximum temperature in the reduction zone around 1000°C, which is better for sponge iron making as accretion in the kiln is not formed within this temperature [10]. It declines the material loss and increase the kiln availability.

7 Conclusions

As the sponge iron industry is passing through several problems, such as lack of proper integration of energy and increased cost of raw materials, in the present work, an effort is made to reduce the energy requirement of the plant in terms of coal consumption. For this purpose, two modifications, case-1 and case-2, are developed, where the heat available with waste gas is integrated in the process to preheat feed material to kiln and air. These cases are then compared with the existing waste heat recovery system of the plant based on coal consumption, operating and capital costs, profit, and payback period. The salient features of the present study are as follows.

1. Among case-1, case-2, and existing waste heat recovery system, case-2 is selected as the best option for the utilization of waste heat. It saves 30.5% of coal in comparison to conventional process. This modification eliminates the use of EC and, consequently, saves maximum of 72.6% water.
2. Sponge iron industry is an energy intensive industry and provides a large scope to decrease its energy bills to make it more competitive. Thus, to reduce energy as well as water consumption in these industries, modification suggested through case-2 can be applied.

Nomenclature

A = heat transfer surface area (m^2)
c = specific heat capacity (J/kg °C)
D = diameter of drier and kiln (m)
d = mean particle diameter (μm)
F = material feed rate in the drier (kg/h)
G = mass flow rate of waste gas in the drier (kg/h)
h = convective heat transfer coefficient used in the drier, duct, and kiln (W/m^2 °C)
k = thermal conductivity (W/m °C)
kW h = kilowatt hour
L = length of the drier and kiln (m)
ℓ = length of the duct (m)
M = mass flow rate (kg/h)
m = mass flow rate of waste gas (kg/s)
N = rotational speed of the drier (rpm)
Nu = Nusselt number, dimensionless
Pr = Prandlt number, dimensionless
Q = heat transfer rate (kJ/h)
Re = Reynold's number, dimensionless
r = radius (m)
T = temperature (°C)
$\overline{T}$ = average temperature (°C)
t = tonne (=1000 kg)
V = volume of the waste gas flowing through the drier (m^3/h)
v = velocity of the waste gas inside the duct and drier (m/s)
x = thickness (m)

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Greek Letters

α = slope of the drier (deg)
λ = latent heat of vaporization (kJ/kg)
σ = density of waste gas flowing inside the drier (kg/m^3)
ν = kinematic viscosity of waste gas inside the duct and drier (m^2/s)
τ = residence time of materials in the drier (min)

Subscripts

a = ambient
c = coal
coa = convective heat transfer coefficient from outer surface of the drier to outside atmosphere ($W/m^2\,°C$)
d = duct
dp = drier plate
f = waste gas
gl = heat transfer coefficient from within the waste gas to the drier plate ($W/m^2\,°C$)
i = inside and inlet
ms = moisture
o = outside and outlet
p = process and particle
r = refractory
roa = radiative heat transfer coefficient from outer surface of drier to outside atmosphere ($W/m^2\,°C$)
s = solids (iron ore) in the kiln

Appendix A: Preheating of Feed Material

A portion of heat of combustion of coal (12% for 1t moisture) is utilized in evaporating moisture in the form of waste gas [18]. For 500 t/day, capacity flow rates of feed coal and iron ore are 15.875 t/h and 30 t/h, respectively. The moisture contents of feed coal and iron ore are 13% and 2%, respectively.

$$\text{moisture in feed coal} = \frac{13 \times 15.875}{100} = 2.06375 \ t/\text{h}$$

$$\text{moisture in iron ore} = \frac{2 \times 30}{100} = 0.6 \ t/\text{h}$$

The total moisture in feed materials is 2.66375 t/h.

From above, it is calculated that 31.965% heat of combustion of coal per hour is required to evaporate the total moisture content of feed materials in the kiln. Therefore, 31.965% of 15.875 t/h feed coal is consumed for moisture removal, which may be saved when the feed material is preheated using waste gas. This implies that 30t iron ore requires 10.8 t/h moisture free coal for further reduction process in the kiln. Thus, 12.414t coal is required for 30t iron ore.

Thus, 33t iron ore requires 11.88t moisture free feed coal per hour for further reduction process.

Let x be the amount of feed coal, which gives 11.88t moisture free coal. Then

$$x - \frac{13x}{100} = 11.88 \Rightarrow 0.87x = 11.88 \Rightarrow x = 13.655$$

and so, 13.655t coal is required to reduce 33t iron ore to iron in the kiln.

Conventional plant kiln loading

$$\text{capacity} = 15.875 + 30 = 45.875 \ t/\text{h} \quad \text{(A1)}$$

For modified plant kiln

$$\text{loading} = 11.88 + [33 - \text{moisture present in the iron ore}]$$
$$= 44.22 \ t/\text{h} \quad \text{(A2)}$$

From Eqs. (A1) and (A2), it is concluded that although iron ore feeding increases by 10% and feed coal decreases by 14%, the kiln loading capacity is still below.

Appendix B: Design of Rotary Drier

The industrial drier requires 2.23×10^9 J/h heat for evaporation of 1t moisture [23]. In the conventional coal based sponge iron plant, 2.66375t moisture is present in feed materials to the rotary kiln, which requires 5.936×10^9 J/h heat to evaporate. Considering 50% efficiency of industrial drier [23], the total required heat is found as 11.87×10^9 J/h, i.e., 32.97×10^5 W. Inlet and outlet temperatures of waste gas to drier are 750°C and 250°C. Average density and specific heat capacity of waste gas are 1.2 kg/m^3 and 1140 J/kg °C, respectively [18]. From above, required flow rate of waste gas is calculated as 18.15 kg/s. The outlet temperature of feed material after moisture removal is assumed as 110°C [23,24].

Enthalpy change of waste gas in the rotary drier=required heat to heat moisture from ambient temperature to 100°C+required heat to vaporize moisture at 100°C+required heat to heat water vapor to outgoing waste gas temperature, i.e., 250°C+required heat to heat feed coal (moisture free) from ambient temperature to the product temperature, i.e., 110°C+required heat to heat iron ore (moisture free) from ambient temperature to the product temperature, i.e., 110°C+heat lost to surroundings through shell of the drier.

The values of these terms are computed, given as follows.

- Enthalpy change of waste gas in the rotary drier=32.97×10^5 W.
- Required heat to heat moisture from ambient temperature to 100°C=mass of the moisture×specific heat of moisture $\times \Delta T = 2.66375 \times 10^3 \times 4200 \times 70 = 2.17 \times 10^5$ W.
- Required heat to vaporize moisture at 100°C =mass of moisture×latent heat of vaporization $= 2.66375 \times 10^3 \times 2260 \times 10^3 = 16.72$ W.
- Required heat to heat water vapor to outgoing waste gas temperature, i.e., 250°C=number of mole of the moisture ×average specific heat capacity of steam$\times \Delta T = [2.66375 \times 10^3/18 \times 10^{-3}] \times 35.4 \times 150 = 2.18 \times 10^5$ W.
- Required heat to heat the feed coal (moisture free) from ambient temperature to the product temperature, i.e., $110°C = 11.88 \times 10^3 \times 1.09 \times 10^3 \times 90 = 3.24 \times 10^5$ W.
- Required heat to heat the iron ore (moisture free) from ambient temperature to the product temperature, i.e., 110°C $= 32.34 \times 10^3 \times 0.45 \times 10^3 \times 90 = 3.64 \times 10^5$ W.

From the above values, amount of heat lost to surroundings through shell of the drier is found to be 5.02×10^5 W. Drop in temperature of waste gas in the drier due to loss of heat through the shell of drier using flow rate, density, and specific heat of waste gas is computed as 25°C. Average drier heat loss surface area is calculated to be 174.3 m^2, as shown in Table 2.

For the design of rotary drier, it is suggested that linear velocity should be around 3 m/s for solids of 5 mm mean particle size and around 0.3–1 m/s for 1 mm solids [15]. The design flow rate of waste gas in drier is calculated as 18.15 kg/s. Considering the inside diameter as 3 m, linear velocity of waste gas comes to around 2.14 m/s. The mean particle size of feed material is around 12 mm, although 8.4% of the feed material sizes are below 5 mm. From above, linear velocity of waste gas as 2.14 m/s is accepted conveniently for 12 mm mean particle diameter.

Appendix C: Economic Analysis

1 Capital Cost

1.1 Cost of Duct From ABC Outlet to Rotary Drier. Length and thickness of the duct of mild steel are 43 m and 6 mm,

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respectively. Its internal diameter is 1.6 m. The weight and density of the duct are $10.24t$ and 7876 kg/m^3 [25]. Considering the cost of mild steel plate as *Rs* 33,000/t, the total cost of $10.24t$ mild steel plate would be *Rs* 3,37,920/-. The manufacturing, welding, and installation costs of duct are taken as *Rs* 30,000/-, *Rs* 12,738/-, and *Rs* 15,000/-, respectively.

The number of insulating bricks used in lining inside the duct are found as 2870 and the cost of one brick is *Rs* 800/-. Therefore, the purchase cost of insulating brick is *Rs* 22,96,000/-. Considering the commissioning charges as *Rs* 5,00,000/-, the total cost for installation of duct is *Rs* 31,91,658/-.

1.2 Cost of Returning Duct (From Rotary Drier to ESP Input). The length, thickness of duct plate (mild steel), and internal diameter of duct are 20 m, 6 mm, and 1.6 m. Its capital cost is computed in a similar manner as for duct from ABC outlet to rotary drier and found as *Rs* 1,45,320/-.

1.3 Cost of Rotary Drier With All Accessories. These are the following market costs taken from different manufacturing units:

- cost of rotary drier with accessories such as girth gears, support rollers, drier mounted instruments, weigh feeders, conveyors, and vibrofeeders=*Rs* 2,30,00,000/-
- cost of three phase induction motor used to drive the drier (capacity: 55 kW)=*Rs* 22,00,000/-
- cost of gear box for rotary drier=*Rs* 31,00,000/-
- installation cost of rotary drier=*Rs* 7,50,000/-
- total cost of rotary drier with all accessories =*Rs* 2,90,50,000/-

1.4 Cost of Induced Direct Fan, Motor, Bearing, Coupling, and Electrical Works.

- cost of induced direct fan (capacity: 60,000 m^3/h) =*Rs* 32,00,000/-
- cost of three phase motor for induced direct fan (capacity: 39 kW)=*Rs* 20,00,000/-
- cost of bearing of induced direct fan=*Rs* 5,50,000/-
- cost of coupling used to couple fan and motor =*Rs* 77,000/-
- approximate installation cost=*Rs* 7,500/-
- approximate cost of electrical works including material cost=*Rs* 26,00,000/-
- total cost for installation induced direct fan, its motor and electrical works=*Rs* 84,34,500/-

1.5 Approximate Volume of Work and Cost.

- concreting of 800 m^3=*Rs* 5,16,000/-
- structural erection of 320t=*Rs* 10,56,000/-
- equipment erection 160t=*Rs* 5,28,000/-
- total=*Rs* 21,00,000/-

Therefore, the total capital cost involved in installing duct and rotary drier is *Rs* 4,29,21,478/-.

2 Operating Cost

2.1 Production and Selling Price of Sponge Iron. The cost analysis of the sponge iron is presented in Table 5, which shows that production cost of 1t of sponge iron is *Rs* 11600/-. The selling prices for different sizes of sponge iron are mentioned in Table 6. It gives average sales price of sponge iron in all sizes as *Rs* 14100/-.

2.2 Increase in Production of Sponge Iron. It is shown in Appendix A that for modified plant, feed rate of iron ore is enhanced from 30 t/h to 33 t/h. Plant availability is assumed to be 90%. Thus, production of sponge iron is increased to a value of 14,463 t/yr. The average sales price and cost of production are

Table 5 Production cost of sponge iron

Input parameters	Unit cost *Rs*/t	Total requirement per ton of sponge iron	Cost *Rs*/tonne of sponge iron
Iron ore	4063	1.6t	6500
Total coal (feed+injection)	2288	1.18t	2700
Dolomite	3571	0.0269t	96
Electricity	*Rs* 2.54/- per kW h	128 kW h	325
Maintenance			230
Fixed cost			1366
Depreciation			400

Table 6 Average selling price of sponge iron

Size (mm)	Percent of production	Normal market price *Rs*/tonne
+3	85	14,300
From −3 to +1	10	12,800
From −3 to +1	10	14,000 (same as lump sponge iron after briquetting)
−1	5	14,300

Rs 14100/t and *Rs* 11600/t of sponge iron, and so, profit for selling 1t sponge iron is *Rs* 2,500/-, which gives total profit of *Rs* 3,61,57,500/-.

2.3 Decrease in Feed Coal Consumption. The coal consumption in the rotary kiln of modified plant is declined from 21.437 t/h to 17.98 t/h. For plant availability of 90%, coal consumption is reduced to a value of 27255 t/yr. Cost of coal, as shown in Table 5, is *Rs* 2288/t, and so, savings due to reduction in coal consumption is *Rs* 623.6 lakh/yr.

For case-1, gain of around *Rs* 623.6 lakh/yr is predicted and investment is found as *Rs* 429.2 lakh, which gives payback period of 8.37 months.

When the intake capacity of the kiln is increased to 33 t/h, coal consumption is also increased to 19.217 t/h. In this case, total profit due to increased production of sponge iron and reduced coal consumption is found as *Rs* 762.1 lakh/yr, which gives payback period of 6.76 months.

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